# Annual Refinery Capacity Report Explanatory Notes

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## *Refinery Capacity Report* Content

We collect data for the *Refinery Capacity Report* tables on our EIA-820 *Annual Refinery Report* form. The form includes data on the consumption of purchased steam, electricity, coal, and natural gas; refinery receipts of crude oil by method of transportation; operable capacity for atmospheric crude oil distillation units and downstream units; and production capacity for selected petroleum products.

## Frame

The respondent frame consists of all operating and idle petroleum refineries, and beginning in 2024, all non-refinery operators of distillation, reforming, cracking, coking, hydrotreating and similar processes located in the 50 States, the District of Columbia, Puerto Rico, the Virgin Islands, Guam, and other U.S. possessions. As of January 1, 2026, the frame included 130 respondents.

We maintain the respondent frame by monitoring responses to the EIA-810, *Monthly Refinery Report* form, and industry publications for changes and developments in the petroleum industry such as refinery sales, mergers, and new operations.

## Data Collection

We send the EIA-820 form to respondents in January. Respondents are required to submit completed forms by February 15 of the current report year via secure file transfer. We monitor the responses and follow up with companies failing to report by February 15.

## Data Processing and Editing

Upon receipt of respondent survey forms, we transform all reported data into a standard format and send it through a prescreening process. This process validates respondent control information. Then, we process the data using generalized edit and imputation procedures. Automated editing procedures check current data for consistency with past data and for internal consistency (for example, totals equal to the sums of the parts).

After we resolve edit failures and perform imputation for nonrespondents or missing data, we produce and review preliminary tables. This review of the aggregated data may result in further review of respondent-level data, which in turn may result in additional editing and imputation. We may also compare aggregated or respondent-level data to third-party sources of data, such as industry journals.

## Imputation and Estimation

We perform imputation for companies that fail to file their form before the publication deadline. When nonresponse occurs, we impute values for these companies from data reported on the most recent year's Form EIA-820 and from data reported on Form EIA-810, *Monthly Refinery Report*, for that company. For most surveyed items, the value we impute for nonrespondents is the value that the company reported on the Form EIA-820 for the most recent year. For three categories of information, however, we also base the imputed value on their data from the Form EIA-810 as follows:

- Part 4: *Refinery Receipts of Crude Oil by Method of Transportation*. We base the imputation methodology for this section on data reported on both the monthly Form EIA-810 and the annual Form EIA-820. We impute annual refinery receipts of domestic and foreign crude oil for a nonrespondent by aggregating the values for the refinery on the monthly survey. We allocate these values to the method of transportation by using the percentages reported for the refinery in the previous year.
- Part 5: *Atmospheric Crude Oil Distillation Capacity*. We collect operable atmospheric crude oil distillation capacity in barrels per calendar day on the monthly Form EIA-810 as of the first day of each month and on the annual Form EIA-820 as of January 1. As part of the editing process for the Form EIA-820, we compare these two values. We contact companies and resolve any discrepancies by the time of publication. We take imputed values for operable atmospheric crude oil distillation capacity in barrels per calendar day directly from the January Form EIA-810. We then derive a barrel per stream day capacity by dividing the reported barrels per calendar day capacity by 0.95.
- Parts 6 and 7: *Downstream Charge Capacity* and *Production Capacity*. We take current year and projected year data for downstream charge capacity and production capacity directly from the previous year's annual report.

# Dissemination

Prior to 2006, we published the data collected on Form EIA-820, *Annual Refinery Report*, in the *Petroleum Supply Annual, Volume 1*. Beginning with data for 2006, we publish the Form EIA-820 data as a standalone product in the *Refinery Capacity Report*. This report contains statistics on consumption of purchased steam, electricity, coal, and natural gas; refinery receipts of crude oil by method of transportation; current and projected capacities for atmospheric crude oil distillation, downstream charge and production capacities.

# Data Quality

## Response rates

In most years, the response rate for the Form EIA-820 is 100%, meaning that we receive a response from every refinery on the frame.

## Non-sampling errors

There are two types of errors usually associated with data produced from a survey: sampling errors and nonsampling errors. Although we refer to the *Annual Refinery Report* as a survey, it is a census of all petroleum refineries. Therefore, we introduce no sampling error in the data presented in this report.

The data, however, are still subject to nonsampling errors. Nonsampling errors can arise from: (1) our inability to obtain data from all companies in the frame and the method we use to account for nonresponses; (2) definitional difficulties or improperly worded questions, which lead to different interpretations by different respondents; (3) mistakes in recording or coding the data obtained from respondents; and (4) other errors of collection, response, coverage, and estimation.

We employ quality control procedures during data collection and editing to minimize misrepresentation and misreporting of data. We also use nonresponse follow-up procedures to reduce the number of nonrespondents, with a 100% response goal. Additionally, the procedures we employ to impute missing data introduce a minimal amount of error, given the relatively high response rates and follow up to reduce instances of missing data on completed forms.

### Resubmissions

We require EIA-820 resubmissions whenever we discover an error greater than 5% of the true value. In the event of a reporting error, we contact the company and request a corrected report resubmission.

### Revision policy

The data we publish in the *Refinery Capacity Report* reflects our final data on refinery capacity. We will revise it only if, in our judgment, a revision is expected to substantively affect understanding of overall U.S. refinery capacity.

## Confidentiality

The information we collect on operable atmospheric crude oil distillation capacity, downstream charge capacity, and production capacity reported on Parts 5, 6, and 7 of Form EIA-820 are not considered confidential, and we publish them at the refinery level in the *Refinery Capacity Report*. We protect all other information reported on Form EIA-820 (for example, Parts 3 and 4 and information that identifies individual respondents) and do not disclose it to the public to the extent that it satisfies the criteria for exemption under the Freedom of Information Act (FOIA), 5 U.S.C. §552, and Department of Energy regulations, 10 C.F.R. §1004.11, implementing the FOIA, and the Trade Secrets Act, 18 U.S.C. §1905.

The Federal Energy Administration Act requires us to provide company-specific data to other federal agencies when requested for official use. We may also make the information reported on this form available, upon request, to another component of the Department of Energy (DOE); to any Committee of Congress, the General Accountability Office, or other federal agencies authorized by law to receive such information. A court of competent jurisdiction may obtain this information in response to an order. We may use the information for any nonstatistical purposes such as administrative, regulatory, law enforcement, or adjudicatory purposes.

We also provide company-specific data to other DOE offices for the purpose of examining specific petroleum operations in the context of emergency response planning and actual emergencies.

We do not apply disclosure limitation procedures to the statistical data published from this survey's information. So, some statistics may be based on data from fewer than three respondents or may be dominated by data from one or two large respondents. In these cases, a knowledgeable person may be able to estimate the information reported by a specific respondent.

# Appendix A. *Refinery Capacity Report* History and Form EIA-820 Updates

- **1918:** The Bureau of Mines, housed in the Department of Commerce, began collecting refinery capacity data on a voluntary basis. In 1980, the data collection was moved to EIA, and we implemented the mandatory Form EIA-177, *Capacity of Petroleum Refineries*. We expanded the form to include information on refining capacity for current year operations, two-year capacity projections, and refinery input and production data. We also added working storage capacity data to the form and product categories for more complete coverage of refining capacity. We expanded data collection on refinery downstream facilities to include a breakdown of thermal operations and to add vacuum distillation, catalytic hydrorefining, and hydrotreating. We also added production capacity to include information on isomerization, alkylation, aromatics, asphalt and road oil, coking, lubricants and hydrogen.
- **1983**: We revised the form to improve the consistency and quality of the data we collected and redesignated it as Form EIA-820, *Annual Refinery Report*. We added two sections for data previously reported monthly: (1) refinery receipts of crude oil by method of transportation and (2) fuels consumed for all purposes at refineries. Also, we eliminated the second-year projections on refining capacity. As a result of research we conducted comparing our motor gasoline data to data collected by the Federal Highway Administration (FHWA), we included motor gasoline blending plants for the first time in the respondent frame to produce more accurate statistics on the production of motor gasoline.
- **1987**: We updated the terminology on the form to reduce respondent burden and to better reflect current refinery operations. We removed sections on projected input and production of refinery processing facilities. We combined several categories under catalytic hydrotreating: naphtha and reformer feeds into a single category, as well as residual fuel oil and *other*. We also combined thermal cracking types, gas oil, and *other* into a single category. We replaced catalytic reforming types, conventional and bi-metallic with low- and high-pressure processing units. We added two new categories: fuels solvent deasphalting to downstream charge capacity and sulfur recovery to production capacity.
- **1994**: We revised the form to enable EIA to calculate utilization rates for certain downstream processing units and to reflect storage capacity of fuels mandated by the *Clean Air Act Amendments of 1990*. Additions to the form included calendar day downstream charge capacity for fluid and delayed coking, catalytic cracking, and catalytic hydrocracking. We also added storage capacity categories for reformulated, oxygenated, and other finished motor gasoline, as well as oxygenate storage capacity and separate categories for high and low sulfur distillate fuel oil.
- **1995**: We removed motor gasoline blending plants from the survey frame since the only section of the form that applied to them was working and shell storage capacity. We also stopped collecting storage capacity from shutdown refineries, eliminating these refineries from the survey frame.
- **1996**: We moved the survey to a biennial schedule (every other year) and renamed it *Biennial Refinery Report*. We did not conduct the survey in 1996 or 1998, meaning that we do not have data for crude oil and petroleum products consumed at refineries during 1995 and 1997. These

data are available from the Form EIA-810, *Monthly Refinery Report*. Refineries continued to submit data for refinery consumption of natural gas, coal, and purchased steam and electricity on Form EIA-820.

- **2000**: We reverted the survey back to an annual schedule.
- **2004**: We changed the survey form to reflect the increasing emphasis on the removal of sulfur from transportation fuels.
- **2009**: We added natural gas used as feedstock for hydrogen plant production and isooctane production capacity to the form.
- **2010**: We updated the survey form to reflect the increasing use of biofuels. We added storage capacities of biomass-based diesel fuel, other renewable diesel fuel, and other renewable fuels to the form. We also added barrels per calendar day capacity of the catalytic reformer; previously, we collected this only on a stream day basis.
- **2011**: We removed storage capacity data from the form. From 2011 to 2024, refineries reported storage capacity on Form EIA-810, *Monthly Refinery Report*.
- **2024**: We updated the frame to include non-refinery operators of distillation, reforming, cracking, coking, hydrotreating, and similar processes.